

Talk at BITS Pilani Hyderabad

Does memory fade away for exchange rate forecasting?

Tanujit Chakraborty

Associate Professor of Statistics & Data Science

Sorbonne University, Abu Dhabi & SCAI Paris

ctanujit@gmail.com

January, 2026

- Some days (calm periods), FX behaves like an ostrich: **fast forgetting**.
- Other days (shock hits) it's an elephant: slow decay of influence (**long memory**).

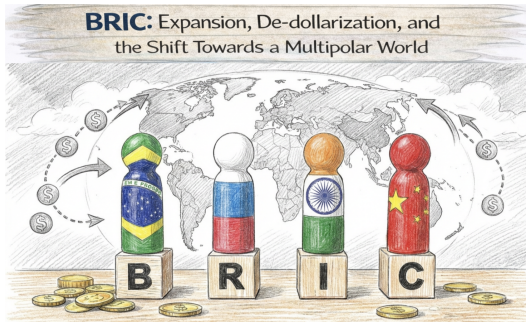


So, when does long memory exist, and how do we exploit it with neural models?

- 1 Exchange Rates
- 2 Time Series Forecasting
- 3 Motivation and Problem Statement
- 4 Methodology
- 5 Theoretical Results
- 6 Empirical Results
- 7 Policy Implications

Exchange Rates

- Exchange rates are central to macroeconomic stability and risk evaluation.
- Understanding the spot exchange rate is important for trade prices, external debt servicing, and international capital movements.
- Emerging economies show stronger volatility than developed markets.
- BRIC (Brazil, Russia, India, and China) economies play an increasingly important role in the global economy.



Exchange Rate Forecasting

- Exchange rate forecasts are critical for monetary policy, capital flow management, and macro-prudential regulation.
- In emerging economies, exchange rate fluctuations strongly influence inflation.
- Spot exchange rates rapidly incorporate new information, making them ideal for analyzing policy shocks and uncertainty.
- For BRIC economies, accurate forecasting supports macroeconomic planning, financial stability, and cross-country policy coordination.



More sophisticated forecasting models required to navigate new and traditional drivers

Accurate exchange rate forecasting for central banks



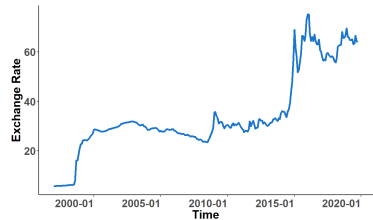
BRIC Exchange Rates



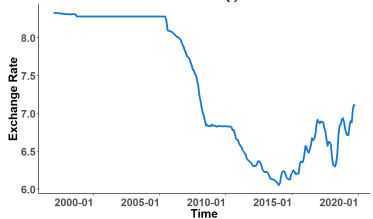
Brazil Exchange Rate



India Exchange Rate



Russia Exchange Rate



China Exchange Rate

Neural ARFIMA model for forecasting BRIC exchange rates with long memory under oil shocks and policy uncertainties

Tanujit Chakraborty^{1,2,*} Donia Beshar¹ Madhurima Panja¹
and Shovon Sengupta¹

¹SAFIR, Sorbonne University Abu Dhabi, UAE and ²Sorbonne Center for Artificial Intelligence, Sorbonne University, Paris, France

*Corresponding author Email: tanujit.chakraborty@sorbonne.ae.

Package 'narfima'

September 21, 2025

Title Neural AutoRegressive Fractionally Integrated Moving Average Model

Version 0.1.0

Description Methods and tools for forecasting univariate time series using the NARFIMA (Neural AutoRegressive Fractionally Integrated Moving Average) model. It combines neural networks with fractional differencing to capture both nonlinear patterns and long-term dependencies. The NARFIMA model supports seasonal adjustment, Box-Cox transformations, optional exogenous variables, and the computation of prediction intervals. In addition to the NARFIMA model, this package provides alternative forecasting models including NARIMA (Neural ARIMA), NBSTS (Neural Bayesian Structural Time Series), and NNaive (Neural Naive) for performance comparison across different modeling approaches. The methods are based on algorithms introduced by Chakraborty et al. (2025) <[doi:10.48550/arXiv.2509.06697](https://doi.org/10.48550/arXiv.2509.06697)>.

Maintainer Donia Beshar <donia.a.beshar@gmail.com>

Time Series Forecasting

Definition: A time series is a set of observations, each one being recorded at a specific time. (e.g., Annual GDP of a country, Sales figure, etc).

- **Stationarity:** Statistical properties such as mean and variance remain constant over time.
- **Autocorrelation:** Measures the correlation between observations at different lags.
- **Forecasting** is the process of estimating how the sequence of observations will continue into the future.

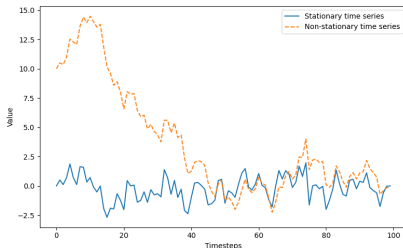


Figure: Visualization of a simulated stationary and non-stationary time series

Data Characteristics: Exchange Rate Series

- Exchange rate time series exhibit long memory dynamics and nonlinear characteristics.

Table: Characteristics of exchange rate Series for BRIC countries.

Country	Properties			
	Linearity	Stationarity	Seasonality	Long Memory (Hurst exponent)
Brazil	×	×	×	✓(0.784)
Russia	×	×	×	✓(0.824)
India	✓	×	×	✓(0.848)
China	×	×	×	✓(0.868)

Conclusion: ✓ Detected × Not detected

Short Memory

- Short-memory (or short-range dependent) time series are data sequences where the correlation between observations decays rapidly—usually exponentially—as the time lag increases, typically allowing for swift convergence to a mean.
- They are characterized by finite variance, differ from long-memory processes, and are often modeled using AR, MA, or ARMA frameworks.
- Key Characteristics and Analysis:
 - **Rapid Correlation Decay:** The autocorrelation function (ACF) of a short-memory series drops quickly towards zero.
 - **Stationarity:** These series are often stationary, meaning their statistical properties (mean, variance) do not change over time.
 - **Memory Type:** Unlike long-memory series (which have long-range dependencies), short-memory series depend primarily on recent observations.

- Consider a stochastic process $\{Y_t\}$ with mean μ and variance σ^2 . To study how past values influence the present, we look at autocovariance

$$\gamma(k) = \mathbb{E}[(Y_t - \mu)(Y_{t-k} - \mu)].$$

- In short memory processes, $\gamma(k)$ decreases rapidly as k increases.
- In long-memory processes, the autocorrelation

$$\rho(k) = \frac{\gamma(k)}{\gamma(0)} = \frac{\gamma(k)}{\sigma^2}$$

decays slowly as $\rho(k) \sim ck^{2d-1}$, with $d \in (0, \frac{1}{2})$ and $c > 0$.

- Equivalently, the autocovariance sum diverges: $\sum_{k=0}^{\infty} \gamma(k) = \infty$.

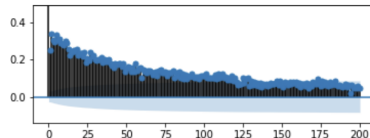
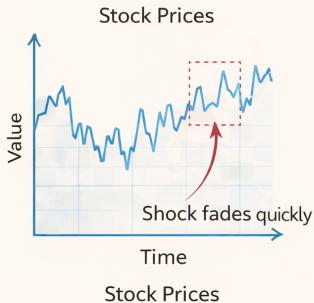


Fig: ACF plot of daily DOW Jones closing prices
(Zhao et al. ICML 2020).

Economic Time Series:

Short Memory vs Long Memory

Short Memory

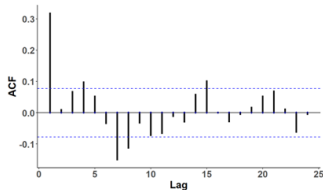


Long Memory

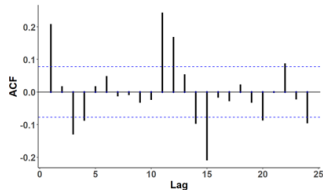


Exchange Rates and Long Memory

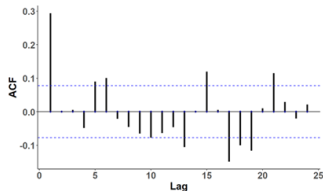
Brazil



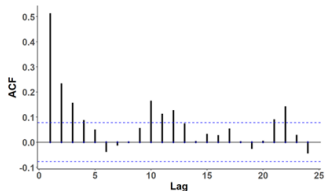
Russia



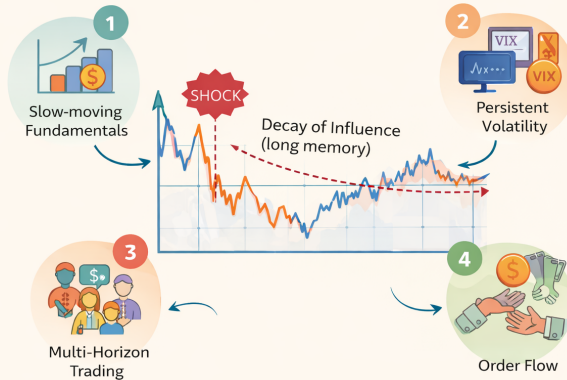
India



China



Long Memory in Exchange Rates: An Economic Interpretation



Fractional Differentiation

- Time series data with high persistence, serial correlation, and nonstationarity pose a significant challenge for time series forecasting models.
- A common approach is to perform integer-order differencing to induce stationarity.
- First- and second-order differencing remove trends but also eliminate long-term memory in the data.
- **Brief history:** Leibniz first suggested fractional derivatives in the 17th century, blending integration and differentiation, allowing nonstationarity removal while retaining long-term memory.

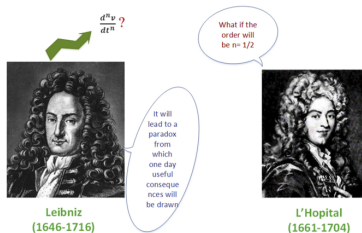


Fig: Concept of fractional calculus can be traced back to Leibniz's letter to L'Hopital in 1695.

- [Granger 1980 and Hosking 1981](#) introduced fractional non-integer differencing as a way to handle long memory by controlling the persistence of a time series through the operator $(1 - B)^d$, where $B Y_t = Y_{t-1}$.
- Expanding $(1 - B)^d$ using an infinite binomial series gives

$$(1 - B)^d Y_t = \sum_{k=0}^{\infty} (-B)^k Y_t \prod_{i=0}^{k-1} \frac{d-i}{k-i} = \sum_{k=0}^{\infty} w_k Y_{t-k}.$$

- An iterative formula for the weights of the lags

$$w_k = -w_{k-1} \cdot \frac{d - k + 1}{k}.$$

- Many macroeconomic time series (e.g., exchange rate of emerging economies) combine two key properties: **nonlinearity and long-range dependence**.
- Classical ARFIMA models capture long memory but remain restricted to linear dynamics.
- Modern **neural networks** learn complex nonlinear patterns but do not preserve long-range dependence.
- How to design a **nonlinear neural forecaster that perceives long memory?**

Proposal: Neural ARFIMA Model

- NARFIMA models the complex dynamics of exchange rate series by combining:
 - 1 Linear long-memory dependence via fractional differencing.
 - 2 Nonlinear effects using a feedforward neural network.
- Let $\{y_t\}_{t=1}^T$ be a univariate time series and $\{X_{j,t}\}_{t=1}^T, j = 1, \dots, r$ be r exogenous covariates.
- The ARFIMAx($\tilde{p}, d, \tilde{q}$) model ([Granger and Joyeux, 1980](#)) is defined as:

$$\underbrace{\left(1 - \sum_{i=1}^{\tilde{p}} \tilde{\phi}_i B^i\right)}_{AR(\tilde{p})} (1 - B)^d y_t = \tilde{\mu} + \underbrace{\sum_{j=1}^r \tilde{\pi}_j B X_{j,t}}_{Covariates} + \underbrace{\left(1 + \sum_{k=1}^{\tilde{q}} \tilde{\theta}_k B^k\right)}_{MA(\tilde{q})} \epsilon_t. \quad (1)$$

- The ARFIMAx residuals, $e_t = y_t - \hat{y}_t^{ARFIMAx}$, contain unexplained patterns by the linear ARFIMAx model.
- Teräsvirta neural network test detects the nonlinear dynamics of e_t .

NARFIMA Model Formulation

- To model nonlinear dependencies, we feed p lagged values of the time series y_t , q lagged values of ARFIMAX residuals e_t , and one lagged value for each of the r covariates into a single hidden-layer feedforward network, optionally including skip connections (inputs directly influence the output and preserve linear relationships).
- The output of the network with skip connections is the one-step-ahead forecast of the series generated as:

$$\begin{aligned} \hat{y}_{t+1} = & \mu_0 + \sum_{l=1}^k \mu_l \sigma \left(\tilde{\alpha}_l + \sum_{i=1}^p \beta_{i,l} y_{t-i+1} + \sum_{j=1}^q \gamma_{j,l} e_{t-j+1} + \sum_{m=1}^r \delta_{m,l} X_{m,t} \right) \\ & + \sum_{u=1}^p \phi_u y_{t-u+1} + \sum_{v=1}^q \eta_v e_{t-v+1} + \sum_{w=1}^r \zeta_w X_{w,t}, \end{aligned}$$

- Network weights and biases are estimated using gradient descent backpropagation ([Rumelhart and Hinton, 1986](#)).
- Hyperparameters (p, q, k, skip) are selected via cross-validation to minimize RMSE on the validation set.

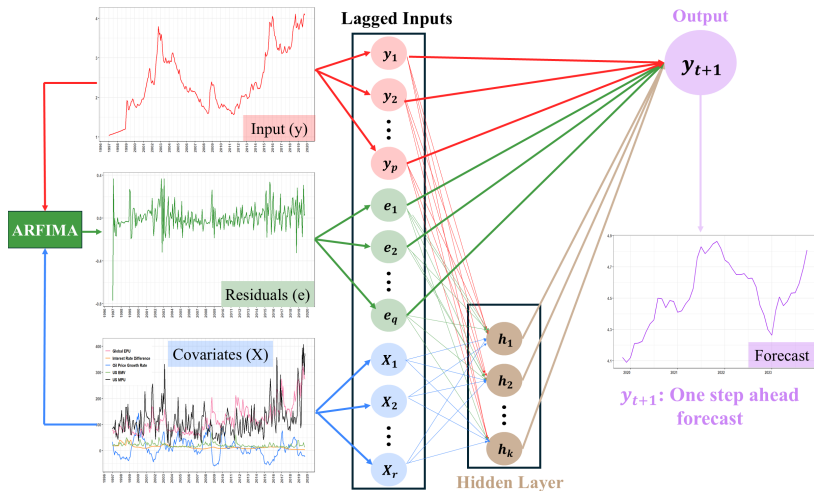


Fig: Architecture of the NARFIMA model.

- We study the underlying random processes.
- The paper in which 'Markov chains' first make an appearance in his writings (Markov, 1906) concludes with the sentence... "Thus, independence of quantities does not constitute a necessary condition for the existence of the law of large numbers."
- Markov chains are dependent, and Markov proved the law of large numbers and the central limit theorem for Markov chains. Therefore, free will is not necessary to do probability.

Definition (Ergodicity, Meyn and Tweedie, 1996)

A Markov chain $\{x_t\}$ is geometrically ergodic if there exists a probability measure Π on a probability triple $(\mathcal{X}, \mathcal{B}, \lambda)$ and a constant $\rho > 1$ such that $\lim_{n \rightarrow \infty} \rho^n \|\mathbb{P}^n(x, \cdot) - \Pi(\cdot)\| = 0$ for each $x \in \mathcal{X}$ and $\|\cdot\|$ denotes the total variation norm. Then, we say the distribution of $\{x_t\}$ converges to Π and $\{x_t\}$ is asymptotically stationary.

To establish geometric ergodicity of NARFIMA, we follow these steps:

- 1 Establish **irreducibility** of the Markov chain.

Definition (Irreducibility, Meyn and Tweedie, 2012)

A Markov chain $\{x_t\}$ is irreducible if $\forall x \in \mathcal{X} \exists n \in \mathbb{N}; \mathbb{P}^n(x, A) > 0 \forall A \in \mathcal{B}(\mathcal{X})$ with $\lambda(A) > 0$.

- 2 Verify the **geometric drift condition**.

Geometric Drift Condition: $\mathbb{E}[V(y_t) \mid y_{t-1} = y, e_{t-1} = e] \leq (1 - \delta)V(y) + B$ with $\delta \in (0, 1)$ and $B < \infty$.

Objective: Show that the NARFIMA(p, q, k , skip = TRUE) process converges to a unique stationary distribution.

Assumptions:

- ① ARFIMA residuals are stationary.
- ② Activation function is bounded, nonconstant, and asymptotically constant.
- ③ Controllable linear part: $\sum_{u=1}^p \phi_u + \sum_{v=1}^q \eta_v \neq 0$.
- ④ The distribution of the noise process is absolutely continuous with respect to the Lebesgue measure with a continuous strictly positive density on $\mathcal{R}$. The random noise process is bounded with finite variance.
- ⑤ Autoregressive connection stability: $|\sum_{u=1}^p \phi_u| < 1$.

Lemma (Irreducibility)

Suppose that Assumption 4 holds. Then, the Markov chain associated with the NARFIMA process is λ -irreducible on the state space $(\mathcal{R}, \mathcal{B}(\mathcal{R}))$.

To establish geometric ergodicity of NARFIMA, we follow these steps:

- 1 Establish **irreducibility** of the Markov chain.
- 2 Construct the **Lyapunov function** $V(y) = 1 + y^2$ and verify the **drift condition**.

Geometric Drift Condition: $\mathbb{E}[V(y_t) \mid y_{t-1} = y, e_{t-1} = e] \leq (1 - \delta)V(y) + B$ with $\delta \in (0, 1)$ and $B < \infty$.

Connection to Positive Recurrence:

- By **Foster-Lyapunov theorem**, if a Markov chain is irreducible and satisfies a geometric drift condition, it is **positive recurrent**.
- Positive recurrence implies the existence of a unique invariant distribution and convergence to it at a geometric rate.

Key Insight:

- Bounded, nonconstant activation ensures the system can explore the state space.
- Controllable linear part bounds the influence of past states.
- Noise with finite variance.

$\implies$ The geometric drift condition holds globally.

Theorem (Geometric Ergodicity)

If Assumptions 2–4 are satisfied. Then, Assumptions 1 and 5 are sufficient conditions for the geometric ergodicity of the Markov chain associated with the NARFIMA process.

Why are Assumptions 1 and 5 still needed?

- **Assumption 1 (Stationary ARFIMA residuals):** Ensures that residuals do not introduce explosive behavior. Without it, the residuals could grow unbounded, violating the drift condition.
- **Assumption 5 (Contractive autoregressive connections):** Guarantees that the autoregressive part of the control system is contractive, which is necessary to satisfy the geometric drift condition and hence positive recurrence.

Corollary (Asymptotic Stationarity)

Under the conditions of the theorem, the NARFIMA process is asymptotically stationary.

Practical Significance of Theoretical Results

- Estimated weights in an irreducible NARFIMA model are likely close to true parameters; violations of irreducibility can lead to misspecification.
- These theoretical results guarantee that the proposed NARFIMA model cannot show 'explosive' behavior or growing variance over time.
- Geometric ergodicity ensures rapid convergence to the stationary distribution, enabling accurate inference from a single long-run realization.
- The neural network component captures nonlinearity, asymmetry, and higher-order interactions, allowing reliable modeling of regime shifts, crises, and speculative dynamics.
- Stationarity and ergodicity provide stable and interpretable forecasts, useful for central banks and financial institutions in decision-making under long-memory and nonlinear uncertainties.

Verification of Theoretical Assumptions using BRIC Exchange Rate Data:

- Assumption 1 is valid because the estimated d values for BRIC exchange rate all fall in $(0, 0.5)$, ensuring long-memory but stationary behavior ([Granger and Joyeux, 1980](#)).
- Assumption 2 holds by design: $\sigma(x) = 1/(1 + e^{-x})$ is bounded and asymptotically constant.
- Assumptions 3 and 5 were empirically verified only for the countries and horizons in which the optimal configuration included `skip = TRUE`.
- For horizons where `skip = FALSE`, the model reduces to a variant of the ARNN process, for which asymptotic stationarity have been theoretically established ([Trapletti et al., 2000](#)). Then, the assumptions on the residual process are sufficient to ensure the theoretical properties.

Results: Both Assumptions 3 and 5 were satisfied across all countries and horizons, confirming the theoretical framework.

Causal Drivers of BRIC Exchange Rates

- Exchange rates are influenced by global and country-specific factors.
- Emerging markets are particularly vulnerable to external shocks.
- **Nonlinear Granger Causality** approach is adopted to identify the causal drivers.

Selected Causal Drivers

Driver	Rationale for Inclusion
EPU	Captures economic uncertainty affecting capital flows
US EMV	Measures market risk sentiment
US MPU	Affects currency risk premia with emerging market currencies exhibiting greater sensitivity
Oil Price Growth	BRIC includes both exporters (Russia, Brazil) and importers (China, India)
Short-term IRD	Reflects relative monetary policy stance

Experimental Results: MCB Test

MCB is a nonparametric test that computes the average rank of each forecaster based on its error metric. The model with the minimum average rank is the best.

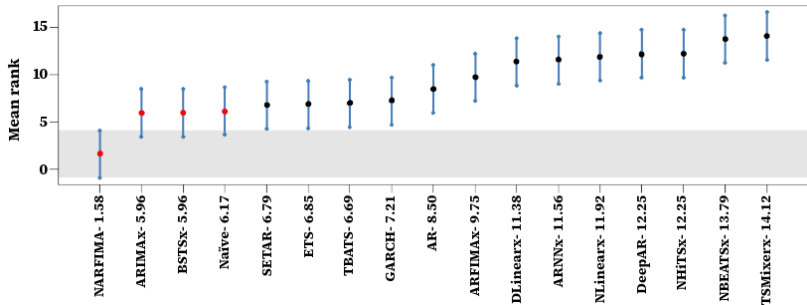


Fig: MCB plot for BRIC based on RMSE metric. Best model determined by the lowest average rank.

Experimental Results: Murphy Diagrams

Murphy diagram technique evaluates forecast dominance across different loss functions. Lower score indicates better performance.

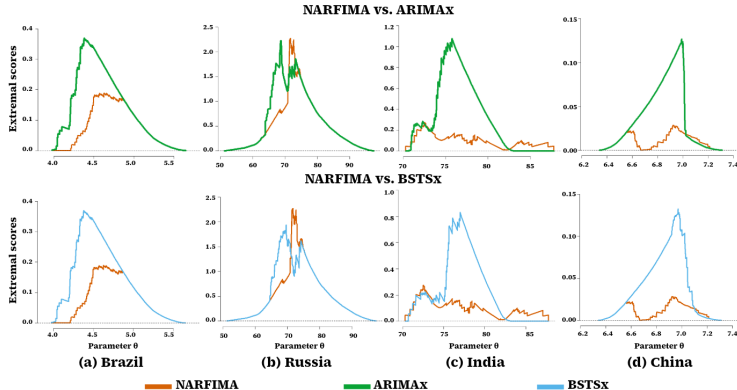


Fig: Murphy diagrams of NARFIMA with baselines ARIMax and BSTSx. Lower scores indicate better performance.

MCB is a nonparametric test that computes the average rank of each forecaster based on its error metric. The model with the minimum average rank is the best.

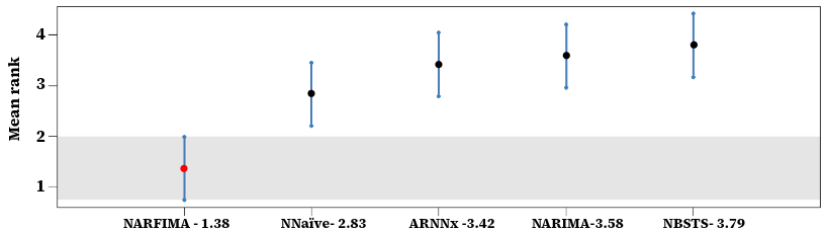


Fig: MCB plot comparing the performance of NARFIMA and its variants based on RMSE metric.

It is a distribution-free approach that converts uncertainty scores into prediction intervals containing the true outcome with a user-specified probability.

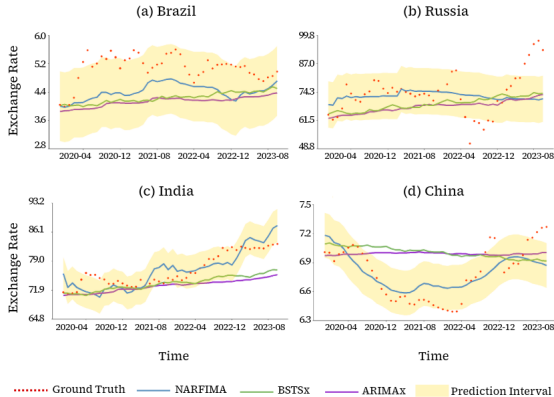
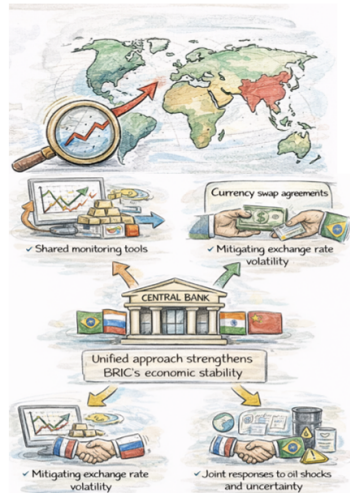


Fig: Exchange rate trajectories of BRIC exchange rates plotted with the 48-month-ahead forecasts and the 95% conformal prediction intervals.

Policy Implications

- Exchange rates are central to macroeconomic stability and monetary policy in BRIC economies.
- De-dollarization adds nonlinear, long-memory dynamics, while partial dollarization sustains USD spillovers.
- Global EPU and US MPU are key causal drivers of BRIC exchange rate volatility.
- Accurate forecasts improves reserve management, FX interventions, and responses to oil and uncertainty shocks.



Limitations of the NARFIMA Model:

- 1 Performance on Linear Series: Traditional models perform competitively.
- 2 Computational Complexity: Two-stage estimation increases computational burden.
- 3 Interpretability: Grey-box nature may limit policy adoption.

Future Research Direction:

- Incorporate additional uncertainty measures such as climate risk indicators and social media-based uncertainty measures.
- Explore alternative neural architectures.
- Extend to other economic and financial applications, including inflation, interest rate, and equity market volatility.



Tanujit Chakraborty
Associate Professor
Sorbonne University



Donia Beshar
Student
Sorbonne University



Madhurima Panja
Statistician
SCAD



Shovon Sengupta
PhD Scholar
BITS